

MVLU COLLEGE

Practical 3: Message Authentication Codes

AIM : Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity.

CODE:

```
# =====  
# Practical : Message Authentication Code  
# Student Name : Sanjana Sahu  
# Class/Roll : TYCST110  
# =====  
  
import hmac  
import hashlib  
  
print("=" * 50)  
print(" MESSAGE AUTHENTICATION CODE (MAC) ")  
print(" Student : Sanjana Sahu")  
print(" Roll No : TYCST110")  
print("=" * 50)  
  
secret_key = input("\nEnter Secret Key : ")  
  
message = input("Enter Original Message : ")  
  
generated_mac = hmac.new(  
    secret_key.encode(),  
    message.encode(),  
    hashlib.sha256  
) .hexdigest()  
  
print("\nGenerated MAC")  
print(generated_mac)  
  
print("\n----- Verification -----")  
  
received_message = input("Enter Received Message : ")
```

```

received_mac = hmac.new(
    secret_key.encode(),
    received_message.encode(),
    hashlib.sha256
).hexdigest()

if hmac.compare_digest(generated_mac, received_mac):

    print("\nMessage Verified Successfully")
    print("Integrity Maintained")
    print("Sender is Authentic")

else:

    print("\nVerification Failed")
    print("Message Modified")

```

OUTPUT:

```

PS C:\Users\IT-02\Desktop\MAC_PRACTICAL 3> & "C:\Program Files\Python314\python.exe" "c:/Users/IT-02/Desktop/MAC_PRACTICAL 3/mac_cli.py"
Generated MAC
c0474e771071014e22e130c112f3e1d1241d0afdd0f1453c0090823f271709b8

----- Verification -----
Enter Received Message : Hello sanjana how are you

Message Verified Successfully
Integrity Maintained
Sender is Authentic
PS C:\Users\IT-02\Desktop\MAC_PRACTICAL 3>

```

GUI:

```

# =====
# Practical : Message Authentication Code
# Student Name : Sanjana Sahu

```

```

# Class/Roll : TYCST110
# =====

import tkinter as tk
from tkinter import messagebox
import hashlib
import hmac

def generate_mac():

    key = key_entry.get()
    message = message_entry.get()

    mac = hmac.new(
        key.encode(),
        message.encode(),
        hashlib.sha256
    ).hexdigest()

    mac_entry.delete(0, tk.END)
    mac_entry.insert(0, mac)

def verify_mac():

    key = key_entry.get()
    original = message_entry.get()
    received = received_entry.get()

    mac1 = hmac.new(
        key.encode(),
        original.encode(),
        hashlib.sha256
    ).hexdigest()

    mac2 = hmac.new(
        key.encode(),
        received.encode(),
        hashlib.sha256

```

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        ).hexdigest()

    if hmac.compare_digest(mac1, mac2):

        messagebox.showinfo(
            "Verification",
            "Message Verified Successfully\n\nStudent : Sanjana Sahu\n\nTYCST110"
        )

    else:

        messagebox.showerror(
            "Verification",
            "Verification Failed\n\nMessage Modified"
        )

root = tk.Tk()

root.title("MAC Authentication - Sanjana Sahu")

root.geometry("550x420")

heading = tk.Label(
    root,
    text="MESSAGE AUTHENTICATION CODE\n\nStudent : Sanjana Sahu (TYCST110)",
    font=("Arial", 14, "bold")
)

heading.pack(pady=10)

tk.Label(root, text="Secret Key").pack()

key_entry = tk.Entry(root, width=40)

key_entry.pack()

tk.Label(root, text="Original Message").pack()

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```
message_entry = tk.Entry(root, width=40)

message_entry.pack()

tk.Button(
    root,
    text="Generate MAC",
    command=generate_mac
).pack(pady=10)

tk.Label(root, text="Generated MAC").pack()

mac_entry = tk.Entry(root, width=65)

mac_entry.pack()

tk.Label(root, text="Received Message").pack()

received_entry = tk.Entry(root, width=40)

received_entry.pack()

tk.Button(
    root,
    text="Verify Message",
    command=verify_mac
).pack(pady=15)

root.mainloop()
```

OUTPUT:

